Lecture Notes: Influence Lines

Lab S2 – Cantilever (Gerber) Bridge Model

SEAA 3012 Civil Engineering Laboratory II
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Learning outcomes

After this session and the laboratory you should be able to

1. define an influence line and distinguish it from a shear-force or bending-moment diagram;
2. derive, by statics alone, the influence lines for the support reactions and hinge forces of the laboratory bridge model;
3. use the Müller–Breslau principle to sketch and check an influence line;
4. use an influence line to find the effect of a single load, a series of loads and a uniformly distributed load, and to locate the critical load position;
5. reduce the dial-gauge readings of the experiment to experimental ordinates, compare them with theory and explain the discrepancies (CLO2, CLO4).

1 Why influence lines?

Most structures are designed for loads that stay where they are: the weight of a floor, the water in a tank. A bridge, a crane girder or a gantry rail carries loads that *move*, and the designer must answer a question that a shear-force diagram cannot: *where* must the vehicle stand to produce the largest reaction at a pier, the largest shear at a section, or the largest moment at midspan? The influence line answers it directly.

Definition

The **influence line** (IL) of a response quantity R (a reaction, a shear force, a bending moment, a deflection) at a *fixed* point of a structure is the graph of that quantity against the position x of a *unit* concentrated load moving across the structure. Its ordinate is written $\eta_R(x)$ and has the units of R per unit load (dimensionless for a reaction).

The contrast with the familiar diagrams is worth stating once and remembering (Table 1).

Table 1: Influence line versus shear-force or bending-moment diagram.

	Shear-force / bending-moment diagram	Influence line
What is fixed	the loads	the point (section, support) where R is evaluated
What varies along the abscissa	the section where V or M is read	the position x of the unit load
What the ordinate means	V or M at that section, for the given loads	R at the fixed point, for a unit load standing at x

For a linear elastic structure the response is proportional to the load and the effects of several loads add (superposition), which is what makes the influence line useful: once $\eta_R(x)$ is known for a unit load, the effect of *any* pattern of concentrated or distributed loads follows by multiplication and addition (Section 5).

2 The laboratory model

The model in the Structures Laboratory (Figure 1) is a cantilever bridge of the type patented by Heinrich Gerber in 1866 and used, at a very different scale, in the Forth Bridge: two anchor spans with cantilever arms, and a *suspended span* that simply rests on the tips of the arms through hinges.

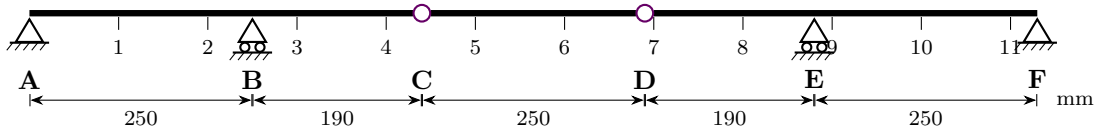


Figure 1: Elevation of the model. Supports A and F are pins, B and E are rollers; C and D are the hinges of the suspended span CD; 1–11 are the halts at which the mass is placed. Dial gauges under A, B, E and F read the support reactions; those under C and D read the hinge forces.

Members. Beam ABC is simply supported at A and B with a cantilever BC of 190 mm. Beam DEF is its mirror image. The suspended span CD (250 mm) rests on the cantilever tips: the hinges at C and D transmit a vertical force but no bending moment.

Supports and hinges. A support or a joint is described by what it *stops*: every movement it prevents produces a force (or moment) in that direction, and every movement it allows produces none.

Restraint	Where	Transmits	Allows
Pin	A, F	vertical and horizontal force	rotation
Roller	B, E	vertical force only	rotation and horizontal movement
Hinge (internal)	C, D	force between the two members it joins	relative rotation, so $M = 0$ there

A hinge is not a support: it joins two members and passes a force from one to the other, but no bending moment. Each hinge therefore adds one equation, $M = 0$ at that point, which is what makes the model statically determinate. In the model the supports at A and F can also pull downwards, and their gauges register uplift as a negative reading.

Determinacy. No horizontal load is applied, so the pins at A and F provide no horizontal reaction and there are four unknown vertical reactions, R_A, R_B, R_E, R_F. Plane statics gives two useful equations for the whole structure ($\sum F_y = 0$ and $\sum M = 0$), and each internal hinge supplies one more ($M = 0$ at C and at D): four equations for four unknowns. The structure is **statically determinate**, and every influence line in these notes follows from equilibrium alone, with no need for EI. That is also why every influence line turns out to be made of straight segments.

Sign convention. Support reactions are positive upwards. The hinge forces R_C and R_D are the forces that the suspended span presses on the cantilever tips (positive downwards on the tip); they are what the gauges at C and D register. Positions are measured from A, $0 \le x \le 1130$ mm.

3 Derivation of the influence lines

The whole derivation uses one idea three times: *take moments about a support.* The order matters: the suspended span is analysed first, because it is the only member whose loads reach the others.

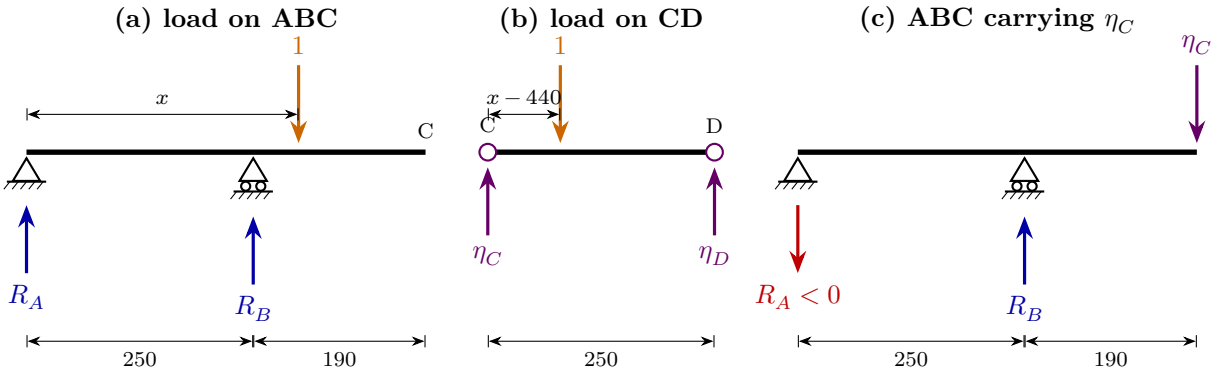


Figure 2: Free-body diagrams used in Steps 1–3. In (c) the pull-down at A is drawn in the direction it actually acts.

Step 1 – Unit load on beam ABC ($0 \le x \le 440$)

Isolate ABC (Figure 2a). It carries the unit load at x and the two reactions R_A and R_B; the hinge at C is unloaded because the suspended span carries nothing. Taking moments about B (anticlockwise positive),

$$R_A \cdot 250 - 1 \cdot (250 - x) = 0 \quad \Longrightarrow \quad \eta_A(x) = \frac{250 - x}{250}, \tag{1}$$

and vertical equilibrium gives

$$\eta_B(x) = 1 - \eta_A(x) = \frac{x}{250}. \tag{2}$$

Nothing in (1) stops at $x = 250$: for a load on the cantilever ($250 < x \le 440$) the lever arm $(250 - x)$ is negative, so η_A is negative (the support at A must hold the beam down) and η_B exceeds 1. At the tip, $\eta_A(440) = -190/250 = -0.76$ and $\eta_B(440) = 440/250 = 1.76$.

Step 2 – Unit load on the suspended span CD ($440 \le x \le 690$)

Isolate CD (Figure 2b). It is a simply supported beam of span 250 mm resting on the hinges, so the hinge forces follow the lever rule:

$$\eta_C(x) = \frac{690 - x}{250}, \qquad \eta_D(x) = \frac{x - 440}{250}, \qquad \eta_C + \eta_D = 1. \tag{3}$$

Step 3 – Carry the hinge force into beam ABC

With the load on CD, beam ABC carries only the downward force η_C at its tip C (Figure 2c). This is Step 1 with the load η_C standing at $x = 440$:

$$\begin{aligned} \eta_A(x) &= \eta_C(x)\,\frac{250 - 440}{250} = -0.76\,\eta_C(x), \\ \eta_B(x) &= \eta_C(x)\,\frac{440}{250} = 1.76\,\eta_C(x), \end{aligned} \qquad 440 \le x \le 690. \tag{4}$$

At C the two expressions for η_A agree (-0.76), so the influence line is continuous; at D, $\eta_C = 0$ and therefore $\eta_A = \eta_B = 0$.

Step 4 – Unit load on beam DEF ($690 \le x \le 1130$)

Beam DEF is the mirror image of ABC about the centre of the model ($x = 565$), with E playing the part of B and F that of A. Reading the geometry from F,

$$\eta_F(x) = \frac{x - 880}{250}, \qquad \eta_E(x) = \frac{1130 - x}{250}, \qquad 690 \le x \le 1130, \tag{5}$$

while for a load on CD the hinge force η_D is carried by DEF exactly as η_C was by ABC: $\eta_E = 1.76\,\eta_D$ and $\eta_F = -0.76\,\eta_D$. A load on ABC or on DEF never reaches the other side, because the suspended span is then unloaded and transmits nothing: $\eta_A = \eta_B = 0$ for $x \ge 690$, and $\eta_E = \eta_F = 0$ for $x \le 440$.

Why η_C and η_D are non-zero only between C and D

The gauge at C reads the force that passes through hinge C, from the suspended span into the tip of cantilever BC. A force can pass through the hinge only when the suspended span itself carries something.

- *Load on ABC* ($x < 440$): ABC stands on its own supports A and B and takes the load to them directly. The suspended span carries nothing, so nothing passes through C: $\eta_C = 0$.
- *Load on CD* ($440 \le x \le 690$): CD is a simple beam between the two hinges, and each hinge takes its lever-rule share, $\eta_C = (690 - x)/250$, from 1 at C to 0 at D.
- *Load on DEF* ($x > 690$): the mirror of the first case; the load goes to E and F, CD stays unloaded and $\eta_C = 0$.

The jump at C. Just left of C the load stands on the cantilever tip, which belongs to ABC, so it goes straight to A and B and $\eta_C = 0$; just right of C it stands on the suspended span at its support, so the whole load passes through the hinge and $\eta_C = 1$. The Müller–Breslau construction (Section 4) gives the same picture: cut the vertical connection at C and lift the end of CD by 1 relative to the tip; ABC does not move, because A and B still support it, while CD rotates about D.

Step 5 – Assemble and check

Table 2 lists the six ordinates at the seventeen halts and Figure 3 plots four of them. Three checks catch most mistakes:

1. *Equilibrium of the whole model:* $\eta_A + \eta_B + \eta_E + \eta_F = 1$ for every x (the unit load must be carried by the four supports).
2. *Continuity at the hinges:* the two formulae for η_A give the same value at C; likewise for η_B, η_E and η_F at C and D.
3. *Slopes:* on ABC the slope of η_A is $-1/250$ per mm; on CD it is $+0.76/250$ per mm. A kink can only occur at a hinge (C, D), never at a support that lies within a rigid member (B).

4 The Müller–Breslau principle

There is a faster way to obtain the *shape* of an influence line, and it is the method practising engineers use to check a computer output.

Müller–Breslau principle

Remove the restraint corresponding to the response quantity R and give the released structure a unit displacement in the direction of R (unit translation for a reaction or shear, unit rotation for a moment). The deflected shape of the structure is the influence line of R.

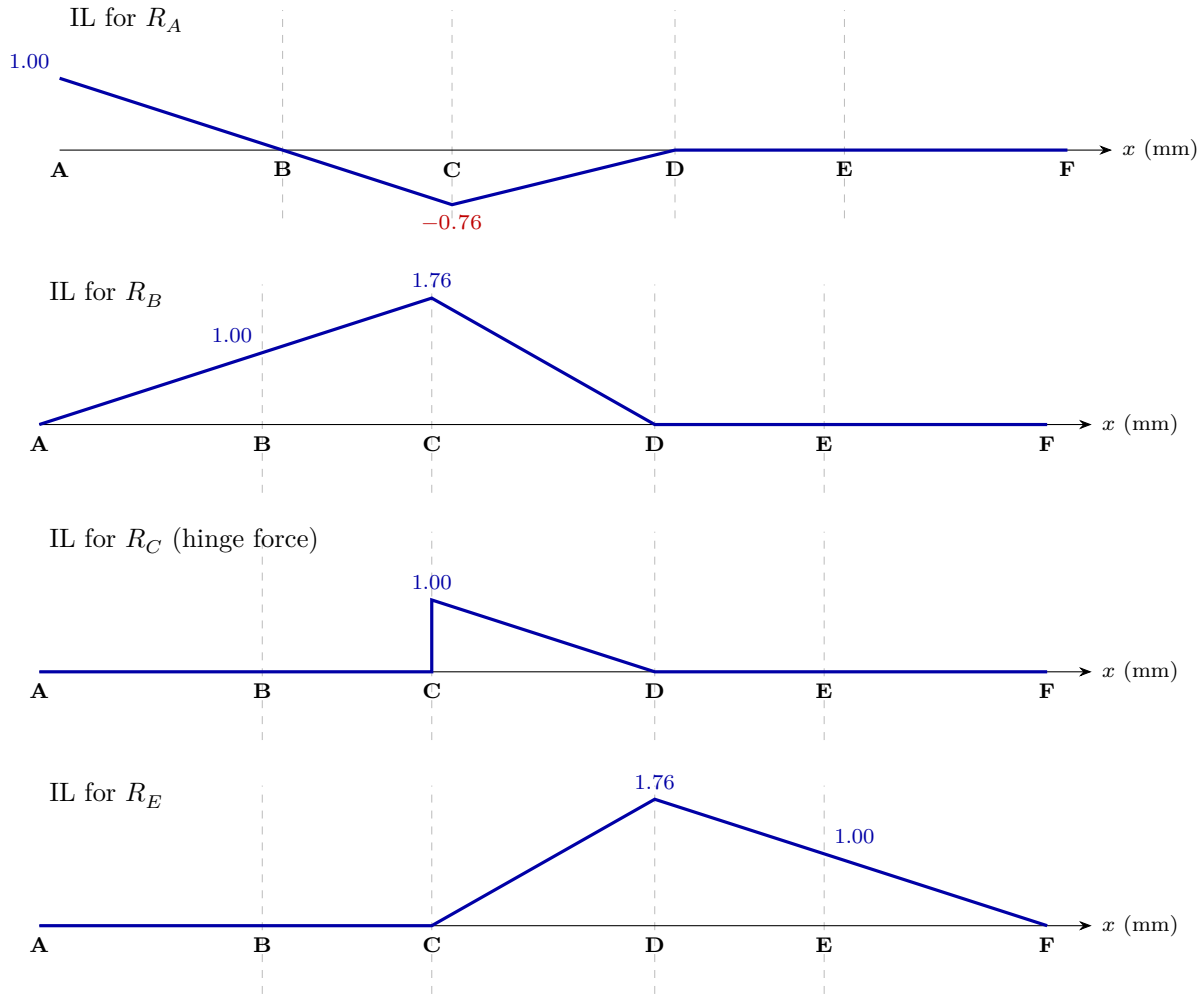


Figure 3: Influence lines for R_A, R_B, the hinge force R_C and R_E (ordinates to scale). R_D and R_F are the mirror images of R_C and R_A about $x = 565$ mm.

Table 2: Theoretical influence-line ordinates (per unit load) at the halts of the model.

Halt	x (mm)	η_A	η_B	η_C	η_D	η_E	η_F
A	0	1.000	0.000	0.000	0.000	0.000	0.000
1	100	0.600	0.400	0.000	0.000	0.000	0.000
2	200	0.200	0.800	0.000	0.000	0.000	0.000
B	250	0.000	1.000	0.000	0.000	0.000	0.000
3	300	-0.200	1.200	0.000	0.000	0.000	0.000
4	400	-0.600	1.600	0.000	0.000	0.000	0.000
C	440	-0.760	1.760	1.000	0.000	0.000	0.000
5	500	-0.578	1.338	0.760	0.240	0.422	-0.182
6	600	-0.274	0.634	0.360	0.640	1.126	-0.486
D	690	0.000	0.000	0.000	1.000	1.760	-0.760
7	700	0.000	0.000	0.000	0.000	1.720	-0.720
8	800	0.000	0.000	0.000	0.000	1.320	-0.320
E	880	0.000	0.000	0.000	0.000	1.000	0.000
9	900	0.000	0.000	0.000	0.000	0.920	0.080
10	1000	0.000	0.000	0.000	0.000	0.520	0.480
11	1100	0.000	0.000	0.000	0.000	0.120	0.880
F	1130	0.000	0.000	0.000	0.000	0.000	1.000

For R_A: remove the support at A and lift that end by one unit. Beam ABC is rigid between the hinge and the support, so it rotates about B as a straight line: A rises by 1, and the tip C, 190 mm on the other side of B, drops by $190/250 = 0.76$. The suspended span follows C and stays straight, so it drops linearly from 0.76 at C to zero at D. Beam DEF is not disturbed at all. The result (Figure 4) is exactly the influence line of Figure 3, including the sign: where the structure moves *against* the direction of the released reaction, the ordinate is negative.

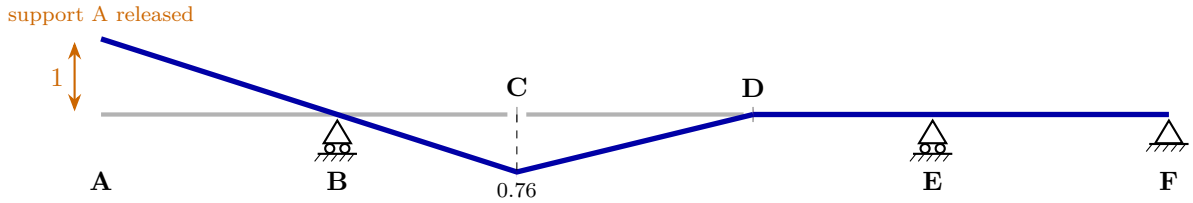


Figure 4: Müller–Breslau construction for R_A: support A released and raised by one unit; the deflected shape (blue) is the influence line.

Two remarks. First, the principle gives the correct *ordinates* as well as the shape only for statically determinate structures, where the released structure is a mechanism made of straight rigid links; for an indeterminate beam the deflected shape is curved and the principle gives the shape qualitatively, the ordinates needing an elastic analysis. Second, the kinks of the deflected shape are at the hinges (C, D) and at the released support, never inside a rigid member. This is a quick visual check of any influence line you compute: Exercise 5 asks you to use it for R_E and R_F.

5 Using an influence line

Let $\eta_R(x)$ be the influence line of R. By superposition,

$$R = P\,\eta_R(x_1) \qquad \text{single load } P \text{ at } x_1, \tag{6}$$

$$R = \sum_i P_i\,\eta_R(x_i) \qquad \text{series of loads } P_i \text{ at } x_i, \tag{7}$$

$$R = \int_a^b w\,\eta_R(x)\,\mathrm{d}x = w \times (\text{area under the IL from } a \text{ to } b) \qquad \text{uniform load } w \text{ on } [a,b]. \tag{8}$$

The quantities in (6)–(8) have whatever units P and w have: with masses in kg the reactions come out in kgf, which is what the laboratory gauges read.

5.1 Critical position of a series of loads

Because every influence line of the model is piecewise linear, the sum in (7) is a piecewise linear function of the position of the load group, and a piecewise linear function attains its extreme values at a vertex. Hence the largest (or most negative) value of R occurs when *one of the loads stands on a vertex of the influence line* (A, B, C or D for R_A). Only a few positions need to be tried.

Worked example 1 – the compound load of Test 2

The 5 kg mass stands at a halt and the 2.5 kg mass 100 mm to its right; the pair moves to the right, so the 2.5 kg mass is in front. With the 5 kg mass at x,

$$R_A(x) = 5\,\eta_A(x) + 2.5\,\eta_A(x+100) \quad \text{(kgf)}.$$

5 kg mass at A: $R_A = 5(1.0)+2.5(0.6) = 6.5$ kgf. *5 kg mass at C:* $R_A = 5(-0.76)+2.5\,\eta_A(540)$, with $\eta_A(540) = -0.76 \times (690-540)/250 = -0.456$, so $R_A = -3.80-1.14 = -4.94$ kgf. *2.5 kg mass at C* (5 kg mass at 340 mm): $R_A = 5\,\eta_A(340)+2.5(-0.76) = 5(-0.36)-1.90 = -3.70$ kgf. The most severe downward pull at A therefore occurs with the *heavier* load on the vertex C. Table 3 gives R_A and R_B at every halt; note that the largest R_B, 11.44 kgf, is more than 1.5 times the total mass of 7.5 kg.

Table 3: Compound load (2.5 kg mass 100 mm to the right of the 5 kg mass): theoretical R_A and R_B in kgf. Exact ordinates were used; the laboratory manual rounds $\eta_A(500)$ to −0.58 and $\eta_A(600)$ to −0.27, which gives −3.58 and −1.35 kgf at halts 5 and 6.

5 kg mass at	x (mm)	2.5 kg at (mm)	R_A (kgf)	R_B (kgf)
A	0	100	6.50	1.00
1	100	200	3.50	4.00
2	200	300	0.50	7.00
B	250	350	-1.00	8.50
3	300	400	-2.50	10.00
4	400	500	-4.44	11.34
C	440	540	-4.94	11.44
5	500	600	-3.57	8.27
6	600	700	-1.37	3.17
D	690	790	0.00	0.00

5.2 Distributed load: area under the influence line

The areas under η_A are, by the triangle rule, $+\frac{1}{2}(250)(1.0) = +125$ mm over AB, $\frac{1}{2}(190)(-0.76) = -72.2$ mm over BC and $\frac{1}{2}(250)(-0.76) = -95.0$ mm over CD, and zero over DEF.

Worked example 2 – a uniformly distributed load

A load of $w = 0.01$ kg/mm (2.5 kg spread over 250 mm) is applied to the model.

- Over AB only: $R_A = 0.01 \times 125 = 1.25$ kgf (half the load, as expected for a simply supported span).
- Over BC and CD: $R_A = 0.01 \times (-72.2 - 95.0) = -1.67$ kgf.
- Over the whole model: $R_A = 0.01 \times (125 - 72.2 - 95.0) = -0.42$ kgf.

The last result says that a bridge of this type loaded uniformly along its whole length *lifts* at A, and the support must be able to hold the deck down. Check by statics: beam ABC carries $w(250+190)$ directly plus half the suspended span, $125w$, at C; moments about B give $R_A \cdot 250 = 250w(125) - 190w(95) - 125w(190) = -10\,550w$, i.e. $R_A = -42.2w$. The influence line gives the answer without redrawing a free-body diagram for each case; that is the point of it.

6 The experiment and its data reduction

What is measured. A 2.5 kg mass is placed at each of the seventeen halts in turn (Test 1), then the 5 kg + 2.5 kg pair (Test 2). At each halt the six dial gauges are read. The gauges register force; with the calibration used in the manual, one division corresponds to 0.05 kgf, so

$$\eta_{\text{test}} = \frac{(\text{reading in divisions}) \times 0.05}{m}, \qquad m = 2.5 \text{ kg (Test 1)}, \tag{9}$$

is the experimental ordinate per unit load, directly comparable with Table 2. For Test 2 the reading $\times\, 0.05$ is the reaction in kgf, comparable with Table 3.

What to plot. For each reaction, the theoretical and experimental influence lines on the same axes, with the halts marked. The experimental points should form the same straight segments; a point that is off the line by more than the error estimate below usually means a misread gauge or a mass that was not centred on the halt.

How to compare. Report the difference $\eta_{\text{test}} - \eta_{\text{theory}}$ at every halt. A percentage difference is only meaningful where the theoretical ordinate is not small: at halt 2, where $\eta_A = 0.2$, a gauge error of one division (0.02 per unit load) is already 10 %, and at B, D and on DEF, where the theoretical value is zero, the percentage is undefined. Judge the agreement by the absolute difference and by the shape of the line, and quote percentages only at the large ordinates (A, C, 4, 5).

Equilibrium check in the laboratory. With the mass at any halt, the four support gauges should add up to the applied mass: $(a+b+e+f) \times 0.05 \approx 2.5$ kgf. A sum that is consistently low points to friction in the gauge mechanisms or a zero error; check it before leaving the laboratory, while the readings can still be repeated.

7 Sources of error

Each source below can be turned into a number; Table 4 does so for the two largest.

Combining the first two in quadrature gives about ± 0.03 per unit load. Against an ordinate of 1.0 at A this is 3 %; against 0.2 at halt 2 it is 15 %. This is the quantitative reason for the advice in Section 6: the *same* measurement quality gives very different percentages, so the percentage alone says little.

8 Beyond the laboratory

Real bridge decks are designed for standard vehicle and lane loads that are placed, by means of influence lines or influence surfaces, in the positions that produce the largest effects at each

Table 4: Error budget for one ordinate of η_A (Test 1).

Source	Estimate	Effect on η_A
Mass placed ±5 mm from the halt	slope $\times \Delta x$ = (1/250)(5) on ABC; (0.76/250)(5) on CD	±0.020; ±0.015
Gauge resolution, ±1 division	0.05/2.5	±0.020
Zero drift between readings	re-zero after each halt; otherwise up to 1–2 divisions	±0.02–0.04
Friction at the hinges and in the gauge	systematic, lowers the sum of the four support readings	check by the sum

section; the load models of the design codes (for example Load Model 1 of EN 1991-2, and the HA/HB loading of BS 5400 used in older Malaysian practice) are defined with this procedure in mind. For continuous (indeterminate) girders the influence lines are curved and are obtained by an elastic analysis, but the Müller–Breslau sketch still tells the engineer which spans to load for the maximum effect: load the span containing the section and every alternate span. The cantilever bridge of the model, with its determinate hinges, is the arrangement that made the long steel spans of the late nineteenth century possible precisely because its forces could be found by statics alone; the price is a deck that lifts at the end supports under some load patterns, as Worked example 2 shows, and anchor supports that must be designed for uplift.

Summary

Key results for the model

- $\eta_A(x) = (250 - x)/250$ on ABC; $= -0.76\,\eta_C(x)$ on CD; $= 0$ on DEF. $\eta_B(x) = x/250$ on ABC; $= 1.76\,\eta_C(x)$ on CD.
- $\eta_C(x) = (690 - x)/250$ and $\eta_D(x) = (x - 440)/250$ on CD, zero elsewhere. η_E, η_F: mirror images of η_B, η_A.
- Check: $\eta_A + \eta_B + \eta_E + \eta_F = 1$ for every x.
- Loads: $R = \sum P_i\,\eta_R(x_i)$; UDL: $R = w\times$ area. The extreme of a series of loads occurs with one load on a vertex.
- Data reduction: $\eta_{\text{test}} = \text{reading} \times 0.05/m$; compare by absolute difference; percentages only at large ordinates.

Activity: Influence Line Explorer (scan the code)

Open the interactive model on your phone. Before each step, write your prediction, then check it.

1. Move a unit load from A to F and watch R_A. Where does it change sign? Where is R_B largest, and why is it larger than the load?
2. Switch to the 5 kg + 2.5 kg pair and find, by trial, the position that gives the most negative R_A. Compare with Worked example 1.
3. Turn on *shade the areas* for R_A and read the three areas; confirm the $-42.2w$ result of Worked example 2.
4. Answer the six prediction questions. Then, in the laboratory, type your gauge readings into the last panel and compare them with theory before you write the report.

Influence Line Explorer

References

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